School of Computer Science Engineering and Technology

Course - B.Tech.
Course Code – CSET-225
Year - 2026

Semester – 5th
Course Name - IMDAI
Semester - ODD

Max. Marks: 4

LAB ASSIGNMENT # 02

CIFAR-10 Image Classification using Convolutional Neural Network and ANN

Objective

To gain hands-on experience in implementing and comparing **Convolutional Neural Networks (CNN)** and **Artificial Neural Networks (ANN/MLP)** for image classification using the CIFAR-10 dataset.

The CNN model must include **three convolutional layers with different kernel sizes**, and performance will be compared against the previously implemented ANN model in terms of classification performance, training efficiency, and parameter count.

You will train and evaluate these models on the CIFAR-10 dataset, analyze performance metrics, visualize predictions, and discuss architectural differences and learning behaviors.

Dataset

Use the CIFAR-10 dataset (10 classes, 32×32 color images). You may use the TensorFlow/Keras built-in loader.

Ensure:

- Images are normalized (e.g., scaled to [0,1]).

- Labels are one-hot encoded.

- Train/test split uses standard CIFAR-10 split (or a validated custom split).

2. Model Implementation

2.1 ANN Model (Baseline)

- Use the same ANN from the previous experiment:
  - Three hidden layers (MLP)
  - Activation function(s) defined by student
  - Flatten input from CIFAR-10

2.2 CNN Model

Design and implement a CNN with the following constraints:

- **Three Convolutional Layers**
  - Each layer must use a **different kernel size** (e.g., 3×3, 5×5, 7×7)
- Include at least:
  - Convolution + Activation
  - MaxPooling (optional but recommended)
  - Flatten + Dense Classification Head
- Other hyperparameters (filters, dropout, optimizers, etc.) are flexible for experimentation

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3. Model Training

For both ANN and CNN:

- Train models for specified epochs (e.g., 50–100)
- Track and store histories for:
  - Training & Validation Accuracy
  - Training & Validation Loss
- Store training time
- Save model weights (optional)

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4. Model Evaluation

For both ANN and CNN:

- Evaluate on test set
- Report:
  - Test Accuracy
  - Test Loss
- Compute and display:
  - **Confusion Matrix**
  - **Classification Metrics**:
    - F1-score
    - Recall
    - Precision
- Display **10 misclassified test samples** with predicted vs true class labels

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5. Comparative Analysis

Compare ANN vs CNN on:

- Accuracy & Loss curves
- Training time
- Parameter count
- Confusion matrices
- Generalization performance
- Effect of spatial feature learning (CNN)
- Limitations of ANN for image data

Present at least:

- Training & validation curves
- Table summarizing:
  - Model
  - Parameters
  - Test Accuracy
  - Training Time

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6. Report

Prepare a concise report including:

- Dataset & preprocessing details
- ANN architecture specification
- CNN architecture specification
- Training procedure
- Metrics & visualizations
- Comparison & interpretation
- Discussion & conclusions

Deliverables

- **Source Code** (Colab/Notebook or Python script)
- **Plots**:
  - Training vs Validation Accuracy
  - Training vs Validation Loss
- **Confusion Matrices**
- **Misclassified samples visualization**
- **Summary Table for Comparison**

- **Short Report or Summary**
- (Optional) Saved Models / Logs

Optional Challenges

- Compare activation functions (ReLU, Sigmoid, Tanh)
- Compare optimizers (Adam vs SGD vs RMSProp)
- Apply data augmentation and measure improvements
- Implement cross-validation (advanced)
- Experiment with Dropout, L2, BatchNorm
- Add explanation on why CNN performs better for vision tasks
- Try deeper CNN or tuning kernel sizes
